MIL-P-45172F(MU)
22 December 1972
SUPERSEDING
MIL-P-45172E(MU)
23 August 1966

MILITARY SPECIFICATION

PROJECTILE, 20MM, API, M53; METAL PARTS

1. SCOPE

1.1 This specification covers the metal parts for Projectile, 20mm, Armor Piercing Incendiary (API), M53.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-651 - Visual Inspection Standards for 20mm Ammunition and Components
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:30Z
Check the source to verify that this is the current version before use.

2.1 (Cont)

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

D7258802	- Cartridge, 20mm, API, M53
D7258804	- Body, Adapter and Rotating Band Assembly
D7258805	- Projectile, 20mm, API, M53; Metal Parts
F7258861	- Carton, for Interplant Shipment of Projectile, 20mm, Assemblies and Metal Parts Assemblies (200)
D7259136	- Shipping Container for Inert Projectile, 20mm, Assemblies and Metal Parts Assemblies (10,000)
D7259192	- Cartridge, 20mm, for Projectile, API, M53, Metal Parts Security Test
IEL 7258805	- Index of Inspection Equipment Lists for: Projectile, 20mm, API, M53

PUBLICATIONS

U.S. ARMY MATERIEL COMMAND

AMCR 715-505 - Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20mm Cartridges

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. -The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)
ASTM B117-64 -- Method of Test for Salt Spray (Fog) Testing
ASTM E18-67 -- Method of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

(Technical Society and Technical Association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:30Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General. -The projectile metal parts shall be in accordance with the requirements of Drawing (Dwg.) D7258805 and with all requirements of the drawings and specifications shown thereon.

3.2 Manufacturing Process. -All parts and assemblies shall be manufactured by a process approved by the contracting officer and no deviation from that process shall be made without his prior approval. (see 6.3)

3.3 Banding. -The rotating band blank and the rotating band seat shall be clean at the time of banding. Shot or sandblasting shall not be used to clean the serrated band seat after knurling. The rotating band blank shall fill the band seat.

3.4 Soundness. -Metal discontinuity in the adapter, of the body, adapter and rotating band assembly, Dwg. D7258804, as determined by eddy current inspection, shall be less than that represented by Production Check Standard (PCS) No. 1.

3.5 Adapter security. -The adapter shall remain secured to the body when subjected to the tension loading specified on Dwg. D7258804.

3.6 Hardness. -The body, adapter and nose shall be in accordance with the respective hardness requirements of Dwgs. D7258804, D7258807 (a detail of D7258804) and C7258809 (a detail of D7258805).

3.7 Protective coating.

3.7.1 Phosphate coating.-The pretreatment process used shall be in accordance with applicable requirements of Dwg. D7258805. The phosphate coating shall be as specified in the requirements of TT-C-490 for appearance and minimum weight, as applicable to the type of phosphate coating.

3.7.2 Paint coating. -The paint coating shall be in accordance with applicable requirements of Dwg. D7258805. Exterior paint adhesion shall be in accordance with applicable requirements of TT-C-490.

3.7.3 Corrosion resistance. -The painted exterior surfaces of the projectile body and adapter shall show no evidence of corrosion (red rust) after exposure to a 5 percent solution of atomized salt spray for 24 hours.

3.8 Metal parts security. -The projectile metal parts assembled into cartridges shall not separate or breakup in the test for metal parts security.

3.9 Workmanship. -The requirements for workmanship are as specified by applicable drawings, reference specifications and the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:30Z
Check the source to verify that this is the current version before use.

3.9.1 Processing defects. -The projectile metal parts shall be free of dents, deep scratches, cuts, sharp edges, burrs, metals flaws, excessive surface roughness and other forms of substandard workmanship.

3.9.2 Cleanliness. -The projectile metal parts shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. -Unless otherwise specified in the contract or Purchase order, the supplier is responsible for the performante of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions .-Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample. -At the beginning of regular production a sample shall be submitted in accordance with contract requirements and shall consist of:

a. Inspection:

50 Rotating band blanks
125 Noses
125 Closure discs
125 Adapters
125 Bodies, unhanded
125 Body-adapter and rotating band assemblies (uncoated)
125 Body-adapter and rotating band assemblies (painted)
3 Standard panels

b. Firing test:

4,000 Body-adapter and rotating band assemblies (painted)
4,000 Noses
4,000 Closure discs

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. Identification shall be in accordance with MIL-STD-1168.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:30Z
Check the source to verify that this is the current version before use.

4.2.1.1 Examination and test. -After inspection and provisional acceptance at source, the sample will be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified by the contract.

4.2.1.2 Initial production sample failure. -Failure of the sample to comply with requirements of the drawings and specifications will result in sample disapproval.

4.3 Quality conformance inspection.

4.3.1 Submission of product. -The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.1 Lot. -A lot shall consist of projectile metal parts produced by one manufacturer in one unchanged process, in accordance with the same drawings and drawing revisions, and same specification and specification revision.

4.3.1.2 Lot identification. -Each projectile assembly, painted, and each packed assembly lot shall be identified in accordance with applicable drawings and MIL-STD-1168. Each packed lot of closure discs and each packed lot of noses shall be identified in accordance with MIL-STD-1169 and MIL-STD-1168.

4.3.2 Examination. -One hundred percent examination shall be performed by the contractor for all critical defects. Examination for major and minor defects shall be performed as specified herein. Each part and each assembly found to be defective shall be rejected.

4.3.2.1 Sampling and acceptance criteria. -The sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235. To determine product acceptability, major or minor defects as listed herein may be considered collectively on a class basis or they may be considered individually. However, where major defects are considered collectively on a class basis, the acceptance number for any individual defect of the class shall be limited by the assigned Acceptable Quality Level (AQL) for individual defect associated with the class. (see 4.3.2.2.c) In addition, where three or less defects appear in a classification, or where MIL-STD-1235 is used, sampling plans applicable to a class shall not be used for major defects. (see 6.4)

4.3.2.2 AQL's. -The following AQL's are assigned to major and minor defects:

a. Individual:

Major	0.25%
Minor	0.40%

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:30Z
Check the source to verify that this is the current version before use.

4.3.2.2 (Cont)

b. Class basis:

Major ---------- 1.50%
Minor ---------- 2.50%

c. Individual defects associated with the class basis:

Major ---------- 0.40%

4.3.3 Classification of defects.-1/ The classification of defects shall be as specified herein. (see 6.5)

4.3.3.1 Rotating band blank (see Dwg. B7258816, covering a detail of Dwg. D7258804).

Categories and Defects	Method of Inspection	
Critical: None defined.		
Major:		
101. Foreign matter	Visual	(see 3.3)
102. Defective metal, including lamination	Visual	
103. Incomplete or deformed	Visual	2/
Minor:		
201. Width, max.	Gage	

4.3.3.2 Body (see Dwg. D7258807, covering detail of Dwg. D7258804).

Categories and Defects	Method of Inspection	
Critical: None defined.		
Major:		
101. Rotating band seat knurling missing or incomplete	Visual	
102. Adapter seat grooving or knurling missing or incomplete	Visual/Gage	3/
103. Diameter, adapter seat after knurling, min.	Gage	
104. Length, base to datum diameter	Gage	
105. Length, datum diameter to adapter seat shoulder	Gage	
106. Concentricity, adapter seat with datum surface	Gage	
107. Concentricity, base cavity with datum surface	Gage	
108. Depth, rotating band seat, knurling, min.	Visual/Gage	4/
109. Diameter, body forward of rotating band seat, min.	Gage	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:30Z
Check the source to verify that this is the current version before use.

4.3.3.2 (Cont)

Categories and Defects	Method of Inspection
Minor:	
201. Length, datum diameter to rear of rotating band seat	Gage
202. Diameter, rotating band seat knurling	Gage
203. Width, rotating band seat	Gage
204. Weight	Balance
205. Flat at nose, max.	Visual
206. Finish	Visual-Feel
207. Width, rotating band seat knurling, min.	Visual/Gage 4/
208. Foreign matter	Visual

4.3.3.3 Adapter (see Dwg. C7258808, covering a detail of Dwg. D7258804).

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Large inside diameter (I.D.), max.	Gage
102. Concentricity, serrated cavity with datum surface	Gage
103. Concentricity, large I.D. with datum surface	Gage
104. Depth, large I.D.	Gage
Minor:	
201. Finish	Visual-Feel
202. Foreign matter	Visual

4.3.3.4 Disc, closure (see Dwg. B7258810, covering a detail of Dwg. D7258805).

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. External profile, except over-all length	Gage
102. Length to datum diameter	Gage
103. Concentricity, datum diameter with datum surface	Gage
104. Concentricity, large O.D. with datum surface	Gage
105. Depth, datum diameter to cavity bottom, min.	Gage
106. Diameter, large O.D.	Gage
107. Diameter, small O.D.	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:30Z
Check the source to verify that this is the current version before use.

4.3.3.4 (Cont)

Categories and Defects	Method of Inspection
Minor:	
201. Over-all length	Gage
202. Finish	Visual-Feel
203. Foreign matter	Visual
204. Depth, datum diameter to cavity bottom, max.	Gage

4.3.3.5 Nose (see Dwg. C7258809, covering a detail of Dwg. D7258805).

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Metal defective	Visual (see 3.9)
102. Inside diameter, straight section	Gage
103. Wall thickness, closed end	Gage
104. Foreign matter on interior	Visual
105. Concentricity, large diameter base end with datum surface	Gage
106. Profile and length, shoulder to open end	Gage
107. Profile and length, shoulder to closed end	Gage
Minor:	
201. Finish	Visual-Feel
202. Foreign matter, other than interior	Visual

4.3.3.6 Body, adapter and rotating band assembly (see Dwg. D7258804).

Categories and Defects	Method of Inspection
Critical:	
1. Ferrous metal discontinuity	Soundness test (see 4.4.4)
Major:	
101. Over-all length, min.	Gage
102. Depth, to body datum diameter	Gage
103. Diameter of bourrelet, min.	Gage
104. Concentricity, body rear of rotating band with datum surface	Gage
105. Concentricity, ogive with datum surface	Gage
106. Concentricity, rotating band with datum surface	Gage
107. Length, rear of rotating band to open end of adapter	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:30Z
Check the source to verify that this is the current version before use.

4.3.3.6 (Cont)

Categories and Defects	Method of Inspection
Major: (Cont)	
108. Profile, crimp groove	Gage
109. Radius, rear edge of crimp groove, max.	Gage
110. Diameter, rotating band	Gage
111. Serrations in adapter missing or incomplete	Visual
112. Location, crimp groove	Gage
113. Inside diameter across serrations	Gage
114. Diameter, body, rear of rotating band, min.	Gage
115. Void at junction of rotating band and projectile body	Visual
Minor:	
201. Finish	Visual-Feel
202. Mouth counterbore missing	Visual
203. Depth and width of undercut, front and rear of rotating band	Gage
204. Base radius missing	Visual
205. Width, rotating band	Gage

4.3.3.7 Body, adapter and rotating band assembly, painted (see Dwg. D7258805).

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Diameter, bourrelet, max.	Gage
102. Diameter, body, rear of rotating band, max.	Gage
103. Weight	Balance
Minor:	
201. Paint not dry	Tactile
202. Marking incorrect, incomplete, illegible, or missing	Visual
203. Paint inadequate	Visual
204. Paint on rotating band	Visual
205. Foreign matter	Visual
206. Rotating band damaged	Visual

1/ MIL-STD-651 shall apply in defining and evaluating visual defects.

2/ A rotating band blank which is deformed or judged to lack sufficient metal in any part of its width or thickness to provide for complete band seat fill and minimum rotating band profile dimensions shall be classed defective.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:30Z
Check the source to verify that this is the current version before use.

3/ Measurement of grooving or knurling of adapter seat for compliance with the minimum depth and width requirements shall apply where visual inspection indicates that the depth or width dimension is borderline.

4/ Measurement of rotating band seat knurling for compliance with the minimum depth and width requirements shall apply where visual inspection indicates that the depth or width dimension is borderline.

4.3.4 Tests. - The tests listed in Table I shall be conducted in compliance with the test methods and procedures prescribed by 4.4. The samples shall be selected in such a manner as to be representative of the entire lot. The sample size and acceptance criteria for each test shall be as specified.

Table I

Test		Sample Size	Item	Rqmt Para
Hardness				
Body	1/	13	Body	3.6
Adapter	2/	80	Body and adapter assembly, uncoated	3.6
Nose	3/	80	Nose	3.6
Adapter security	4/	20	Body, adapter and rotating band assembly, painted	3.5
Band tightness	5/	315	Body, adapter and rotating band assembly, uncoated	3.3
Soundness verification	6/	315	Body, adapter and rotating band assembly, uncoated	3.4
Automatic inspection effectiveness	7/		PCS No. 1 (see 4.4.5)	--
Phosphate coating	8/	3	Standard panels, TT-C-490	3.7.1
Paint adhesion	9/	13	Body, adapter and rotating band assembly, painted	3.7.2
Corrosion resistance	10/	32	Body, adapter and rotating band assembly, painted	3.7.3
Metal parts security	11/	250	Body, adapter and rotating band assembly, painted	3.8

1/ If two or more units of the sample fail to comply with the body hardness requirements, the lot shall be rejected. If one unit fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the hardness requirement, the lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:30Z
Check the source to verify that this is the current version before use.

4.3.4 (Cont)

2/ If two or more units of the sample fail to comply with the adapter hardness requirement, the lot shall be rejected. If one unit fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the hardness requirement, the lot shall be rejected.

3/ If two or more units of the sample fail to comply with the nose hardness requirement, the lot shall be rejected. If one unit fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the nose hardness requirement, the lot shall be rejected.

4/ If one or more units of the sample fail to comply with the adapter security requirement, the lot shall be rejected.

5/ If two or more units of the sample have an average radial void, between the seated band and the band seat, greater than 0.004 inch, the lot shall be rejected. If one unit of the sample has an average radial void, greater than 0.004 inch, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units have an average radial void greater than 0.004 inch, the lot shall be rejected.

6/ Rejection of one or more units of the sample by the flaw detector in the test for adapter metal soundness shall be cause for rejection of the lot.

7/ Failure of the flaw detector to reject PCS No. 1 shall be cause for rejection of all assemblies inspected for “soundness” since the preceding test and shall require requalification of the equipment for automatic inspection. (see 4.4.5)

8/ Failure of the phosphate coating on the standard panels to comply with the applicable requirements shall be cause for rejection of all assemblies phosphate since the preceding test.

9/ If two or more units of the sample fail to comply with the requirement for paint adhesion, the lot shall be rejected. If one unit of the sample fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the requirement for paint adhesion, the lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:30Z
Check the source to verify that this is the current version before use.

4.3.4 (Cont)

10/ If two or more units of the sample fail to comply with the corrosion resistance requirement, the lot shall be rejected. If one unit fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the corrosion resistance requirement, the lot shall be rejected.

11/ The occurrence of one or more of the following casualties, as witnessed by hole(s) in the fragmentation screen(s), or recovery of the part(s) or fragment(s) shall be cause for rejection of the lot, except as otherwise provided:

a. Body-adapter separation or breakup.

b. Loss of nose,

c. Complete rotating band separation. This is defined as detachment of all of the band having band seat knurling imprint.

d. Partial rotating band separation. This is defined as detachment of part of the band having band seat knurling imprint.

e. The following criteria shall apply to thrown band fragments having no evidence of band seat knurling imprint:

(1) The lot will be rejected if three or more assemblies throw band fragments having length not greater than 0.500 inch.

(2) The lot will be rejected if any assembly throws a band fragment having length greater than 0.500 inch (indicative of deep-seated band lamination).

(3) The lot will not be penalized for band particles thrown as a result of normal band fringing or slivering.

4.3.5 Packaging, packing and marking inspection. -Inspection for packaging, packing and marking shall comply with applicable requirements of MIL-STD-1169.

4.3.6 Inspection equipment. -Index of Inspection Equipment List (IEL) 7258805 identifies the applicable inspection equipment lists indicating the equipment required to perform examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL (reference is made to Inspection Equipment List for definition of coding, Dwg. Bl107228, a detail of IEL 7258805).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:30Z
Check the source to verify that this is the current version before use.

4.4 Test methods and procedures.

4.4.1 Hardness. -The method of test shall be in accordance with ASTM E18-67, Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

4.4.1.1 Body. -Each sample body shall be sectioned, prepared and one hardness impression taken within each of the areas as shown on Dwg. D7258807, a detail of D7258804.

4.4.1.2 Adapter. -One hardness impression shall be taken per test assembly in the area prescribed on Dwg. D7258804. No cylindrical correction shall be applied.

4.4.1.3 Nose. -Three hardness impressions shall be taken, approximately 120° apart, on the cylindrical diameter of each nose of the sample. The average of the three readings shall apply as the nose hardness. Test sample noses shall be supported on a mandrel approximately 0.5 inch in diameter. No cylindrical correction shall be applied.

4.4.2 Adapter security. -The test for adapter security shall be conducted using a standard tension testing machine of sufficient capacity. Rate of pull shall be l½ inches ± 1/4 inch per minute.

4.4.3 Band tightness. - The method of test shall be in accordance with all requirements of the operating instructions for the band tightness tester (Dwg. B7259545, a detail of IEL 7258805). The inspection sample shall be cycled through the tester to determine compliance with the rotating band tightness requirement. The total measurement of radial void, registered on the tester indicator at final ram pressure, shall be divided by four (number of indenters) to determine the average radial void.

4.4.4 Soundness. -The method of inspection shall be in accordance with all requirements of Operating Instructions Manual, Eddy Current Flaw Detector, Dwg. D11075203 (a detail of IEL 7258805) for Detecting Metal Discontinuity in Projectile Bodies for 20mm Ammunition, except for those inspection requirements which are reserved for Government application, as specified herein.

4.4.4.1 Application of defect standards. -Master defect standards, referenced in the Operating Instruction Manual, shall apply to resolve inspection discrepancies occurring between the results of defect standards supplied for contractor "soundness" inspection and those supplied for Government "soundness" verification. (see 6.6)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:30Z
Check the source to verify that this is the current version before use.

4.4.5 Automatic eddy current inspection effectiveness. -Equipment checkout shall be performed as stated in the Operation Instruction Manual at the start of each shift and after any adjustment or shutdown. Upon completion of checkout, cycle the PCS #1, front slot and rear slot standards through the inspection system during automatic operation. All three standards must be rejected to assure correct adjustment for reject level and scan limits. During remainder of shift the PCS #1 shall be cycled through the equipment at intervals of 20 ± 5 minutes during automatic operation.

4.4.5.1 Equipment verification. -The Government inspector will maintain independent of the contractor, a set of standards and may cycle these standards tnrough the equipment at any time. Failure of equipment to reject standards will be evidence that the equipment is not properly adjusted and the contractor will be so notified.

4.4.6 Soundness verification. -After equipment checkout and verification as specified in 4.4.4 and 4.4.4.1 the entire lot sample will be inspected with the equipment operating automatically. Any rejected unit of the sample will be automatically cycled through the equipment with the CAL. button pressed (to lower the sensitivity one step below that used for production inspection). Any unit which is rejected at the lower sensitivity shall be classed as a defective.

4.4.7 Phosphate coating. -The method of test shall be as prescribed by TT-C-490 for the type of coating used (see Dwg. D7258805). The prescribed panels shall be processed, with the projectiles to be represented, at the start of daily production and at 4-hour intervals, maximum, thereafter. Appearance and weight of the coating shall be recorded.

4.4.8 Paint adhesion. -The method of test shall be in accordance with TT-C-490. The test shall be made on the painted bourrelet of each unit of the sample forward of the rotating band. Visible exposure of underlying phosphate coating or ferrous metal of any unit of the sample, as a result of the test, shall be recorded as a defect.

4.4.9 Corrosion resistance. -The method of test shall be in accordance with ASTM B117-64. The paint shall not be disturbed, however, the rotating band and unpainted surfaces of the sample projectiles may be masked out by a wax coating or other suitable material. The salt spray solution and exposure time shall be as specified in 3.7.3.

4.4.10 Metal parts security .-The method of test will be in accordance with AMCR 715-505, Volume 8. The test sample metal parts shall be inert charged and assembled into cartridges in accordance with the requirements of Dwg. D7259192, Cartridge, 20mm, for Projectile, API, M53, Metal Parts Test. The test cartridges, at ambient temperature, will be fired in bursts of 50 shots in an M39 gun. The gun barrel will be at ambient temperature for the firing of each burst. The cyclic rate of fire will be recorded for informational purposes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:30Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packing, Level C.-Each lot of projectiles shall be packed as specified by Dwg. F7258861 or Dwg. D7259136. Each lot of closure discs and each lot of noses shall be bulk packed in accordance with MIL-STD-1169.

5.2 Marking and labeling. -Marking or labeling of packed containers shall comply with applicable requirements of MIL-STD-1169.

6. NOTES

6.1 Ordering data. -Invitations for bids and contracts or orders will specify the following:

6.1.1 Title, number, and date of this specification.

6.1.2 Place of inspection, if not place of manufacture.

6.1.3 Packing and marking instructions. (see 5.1 and 5.2)

6.1.4 Provision for shipment and testing of production lot samples for metal parts security. (see 4.4.10)

6.1.5 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.2 First article sample. -Where an initial production sample is required by the contract, the contracting officer will notify the test facility when to proceed with the metal parts security test. (see 4.2) Additional projectiles may be required by the test facility.

6.3 Process deviation. -A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which alters the metallurgical or physical properties of the item of contract.

6.4 AQL's. -The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitation, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:30Z
Check the source to verify that this is the current version before use.

6.5 Intermediate Point inspection. -The classification of defects identifies the defect characteristics (among other things) for acceptance inspection. It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for defect characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which could be detrimental to the attainment of optimum product quality in the end item.

6.6 Defect standards. -The initial supply of defect standards and instructions for contractor and Government application to the eddy current method of inspection, will be furnished with the inspection equipment. Replacement standards shall be requisitioned through normal channels by the Government agency having contract administration responsibility.

NOTE: Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1305-A704

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:30Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 202610-22T02:30Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
Department of the Army

OFFICIAL BUSINESS

Commanding Officer
Frankford Arsenal
ATTN: SMUFA-J4000
Philadelphia, Pa. 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:30Z
Check the source to verify that this is the current version before use.